AI-empowered IoT Security for Smart Cities

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Smart cities fully utilize the new generation of Internet of Things (IoT) technology in the process of urban informatization to optimize the urban management and service. However, in the IoT system, while information exchange and communication, wireless sensor network devices may not be able to resist all forms of attacks, which may lead to security issues such as user data disclosure. Aiming at the information security risks in smart city, the typical technologies in IoT is analyzed from the perspective of IoT perception layer and provides corresponding security solutions for the existing security threats. Regarding the communication security, the emerging wireless technology, long range (LoRa), is discussed, and the performance of wireless communication protocol is analyzed through simulation experiments, to verify that the IoT technology based on LoRa communication technology can improve the security of the system in the construction of smart city. The results show that REBEB, a new backoff algorithm, is similar to the binary exponential backoff algorithm in terms of throughput performance. REBEB focuses more on fairness, which is up to 0.985, and to a certain extent, its security is significantly improved. The fairness of REBEB algorithm is more than 0.4 in different nodes and competing windows, and the fairness of the system is better when the number of nodes is small. To sum up, the IoT system based on LoRa communication can effectively improve the security performance of the system in the construction of smart city and avoid the security threats in the IoT signal transmission.

CCS Concepts: • **Computing methodologies → Distributed artificial intelligence**; • **Networks → Mobile and wireless security**;

Additional Key Words and Phrases: Smart city, internet of things security, artificial intelligence, LoRa communication, REBEB

ACM Reference format:
Zhihan Lv, Liang Qiao, Amit Kumar Singh, and Qingjun Wang. 2021. AI-empowered IoT Security for Smart Cities. *ACM Trans. Internet Technol.* 21, 4, Article 99 (July 2021), 20 pages.
https://doi.org/10.1145/3406115

This work was supported by National Natural Science Foundation of China (No. 61902203) and Key Research and Development Plan—Major Scientific and Technological Innovation Projects of ShanDong Province (2019JZZY020101).
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1533-5399/2021/07-ART99 $15.00
https://doi.org/10.1145/3406115

1 INTRODUCTION

In the global scope, the construction of the smart city is rising in Europe and the United States. The significance of its emergence is to promote the intelligent transformation of urban planning, construction, management, and services by using information technology such as the **Internet of Things (IoT)**, cloud computing, big data, and geographic information systems [1, 2]. For a long time, the smart city has been developing in two ways. First, there is the application of information technology in then urban development, that is, the proposal and deepening of a digital city. Second, there is the development of the urbanization process. The continuous improvement of the urbanization rate makes the problems facing the city increasingly serious, and new concepts such as sustainable development and ecological city are focused on in Reference [3]. As the influence of information technology on urban system adaptation and resource optimization becomes greater, the concept of the smart city emerges as the times require. In the development of the smart city, the driving factors of information technology are relatively large. Most countries and cities emphasize building new information infrastructure and using new technology tools to solve the urbanization problem. The focus of smart city construction in this period is new infrastructure such as internet and mobile communication network [4]. Its intelligent urban services include police early warning, real-time query of public transport information, intelligent traffic signal, intelligent parking, and telemedicine, which improve the quality of life of residents while promoting the optimization of the urban spatial structure.

The construction of the smart city is a complex multi-level and systematic project, including four parts: perception layer, communication layer, support layer, and application layer. **Among them, the perception layer based on IoT is the foundation of a series of activities in smart cities. Through sensor technology, radio frequency identification (RFID)**, embedded system technology, and other sensing devices in IoT technology, information collection and monitoring of infrastructure and environment are realized [5–7]. The communication layer can be regarded as a "three-network integration" communication information platform, which summarizes the information of intelligent buildings, smart homes, and the internet. The purpose of the support layer is to achieve the integration of multi-source data information and lay the foundation for the specific application. The application layer directly provides services for urban residents on the basis of data informatization and intelligence. In the four-tier structure of a smart city, the most critical and complex is the IoT perception layer, which can be connected with sensors through some devices and can realize intelligent control of large-scale things through advanced feedback automation technologies such as cloud computing and intelligent recognition [8, 9].

Through using information collection equipment, IoT connects any object with the internet according to the agreed protocol and carries out information exchange and communication, thus realizing intelligent positioning, identification, tracking, monitoring, and management [10]. IoT brings convenience to people's production and life but also has some defects. At present, the standard specifications of many key technologies in IoT are still in the initial stage. **Wireless sensor network (WSN)** devices may not be able to resist all kinds of attacks, and personal location, finance, and other private information are at risk of being leaked [11]. As the number of IoT terminals increase, the security risks are greater. Because it goes into many fields and affects people's lives in all aspects, the corresponding security problems will also bring serious threats. The existing security technology cannot solve the security problems such as system loopholes and data leakage caused by IoT technology. The security architecture of IoT cannot simply copy the security architecture of the traditional internet. Hence, it is necessary to develop a comprehensive security protection system based on the original security strategy.

At present, there are several studies on security ontology modeling. In different studies, security ontology covers different categories, some focus on basic concepts, and some emphasize specific fields. Building security ontology including assets, vulnerabilities, alerts, threats, security mechanisms, and relationships can help security managers manage risks. Some scholars choose neural network as the algorithm model to solve network anomaly detection and propose Kitsune, a plug-and-play NIDS that can learn to detect attacks on local network in an efficient online way without supervision [12]. Mozzaquatro et al. designed an ontology-based security framework using the knowledge reasoning function of a knowledge map to provide corresponding security services for the discovered threats [13]. However, in related fields, there are few studies on the application of wireless technology to the information security of IoT from the perspective of communication, which will be focused on here.

To summarize, this study mainly analyzes the information security risks in smart city from IoT perception layer, explores the typical technologies in IoT, and provides corresponding security solutions for the existing security threats. This study discusses the new wireless technology, **long range (LoRa)**, from communication security, and analyzes the performance of wireless communication protocol through simulation experiments. It is proved that the IoT technology based on LoRa communication technology can improve the security of the system in the construction of smart city.

2 EXPERIMENTAL AND COMPUTATIONAL DETAILS

2.1 Smart City Architecture Based on IoT

International Business Machines Corporation proposed that IoT is a smart earth, so it is generally believed that IoT has created a smart city. The foundation for constructing a smart city is a communication information platform with three-network integration. On this platform, there are shared resources to realize the convergence of three types of information: smart building, smart home, and IoT [14]. Only when the gathering points of three information communicate, transmit, and share through the three-network integration platform can a city be "smart." The purpose of the three-network integration is to provide each intelligent unit and information gathering point with higher speed information services.

Among them, the realization of building intelligence can be divided into three parts: (1) information communication, which is mainly reflected in the communication among people; (2) equipment management (building automation), through a system for real-time monitoring, control, and management of equipment in the building, which embodies the relationship between objects, and information communication between objects and people; and (3) the security system, which is relatively independent, each part of the division of labor in the building is different, but the purpose is to create a safe, comfortable, and efficient environment for human beings. For the building complex, these devices can be managed and monitored remotely and locally, which is mainly reflected in the concept of IoT management in a fixed situation [15]. The biggest point of a smart home is to set up a smart home terminal, through which the home information and external information are connected to each other, playing the role of a gateway.

The construction of intelligent visual IoT can make intelligent perception and automatic data collection for the elements of urban complex, covering all aspects of the urban complex, such as residential, catering, business, office, entertainment, traffic control, security, and fire control monitoring, lighting, environment, and hydrological detection. The collected data can be visualized and standardized so that managers can conduct management of visual urban complex [16]. IoT is mainly divided into three layers: perception layer, network layer, and application layer. First, the

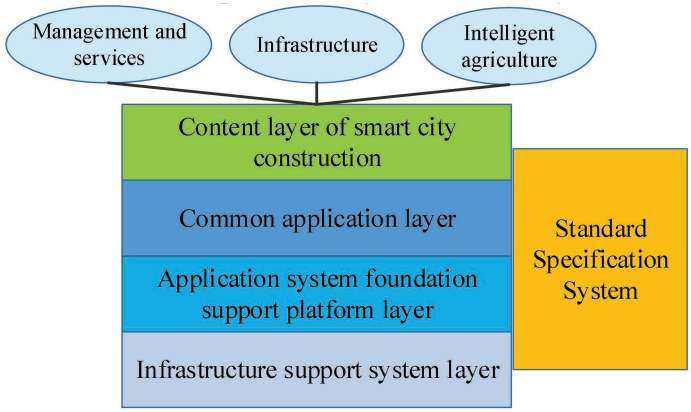

Fig. 1. Overall architecture of smart city system.

perception layer realizes the perception of the pre-set information of the object, that is, the information source of the object itself. Second, it perceives the environment of the object and judges the change of the environment. Third, it perceives the address code of the object, that is, the location of the object. Finally, it perceives the change of the path and the entity of the object in the process of change. Regarding the perception layer of the IoT module, the most important is the application of electronic labels [17]. The information of electronic tag is transmitted by wireless way, and it is transmitted to the perception center through the IoT self-network or three-network integration. Analyzing the system composition and technical function, the perception layer mainly consists of different sensor nodes and field control node equipment, which is used to realize the collection of sensor parameters and the execution of control instructions. The network layer of IoT is mainly composed of ZigBee gateway and router of each subsystem to realize the access of sensing layer nodes and transparent data transmission through ZigBee and TCP/IP [18]. The application layer of IoT can be embodied in smart building, smart home, and smart community. In terms of system composition and technical functions, the application layer consists of management server of application system, web operation platform, and other parts, which mainly realize the management, storage, release, and browsing of system information in a smart city. The overall architecture of the smart city system is shown in Figure 1, including the infrastructure support system layer, basic support platform system layer, smart city construction content layer, and business standard specification system of a smart city [19, 20].

Information and communication technology is the foundation of a smart city. With the emergence and popularization of the internet, the comprehensive coverage and transmission capacity of communication network are improved, as well as the low-cost and functional diversification of communication terminals. The information transmission, calculation, and reception capacity of the city has an explosive growth, and the digital city is born and evolving at a rapid rate. The construction and improvement of application system is an important sign of urban functional informatization. In this stage, all aspects of urban subjects use information technology to complete certain functions, which is embodied in the informatization of production, finance, procurement, and other functions of enterprises, as well as the informatization of business of government commissions and bureaus. In this stage, affairs processing is emphasized, which greatly improves the internal efficiency of urban subjects and accumulates rich business information [21]. The integration of data, management, and service for management and service objects in various fields of the city marks a new stage in the development of smart city. It can organically combine all kinds of information, and the government, citizens, and urban infrastructure receive management and

service. The real smart city system is the real-time, active, and flexible adjustment and optimization of urban ecosystem management and service objects and related elements, thereby achieving the effective balance of demand and supply of resources and services.

2.2 Security Threats in IoT

2.2.1 Internal Threat of IoT. Smart city more obviously impacts various fields of people's life, and smart devices in families, enterprises, communities, and cities are becoming more popular, whereas, in this process, the emergence and application of some cheap networking devices bring new security problems to the whole family and society. Since companies are dramatically increasing the number of sensors and remote monitors that are used to manage the company's office environment and detailed manufacturing processes at the factory. Meanwhile, the government also began to participate in constructing smart cities and hope to use new technologies to improve energy efficiency, reduce traffic congestion, and improve air and water quality. The number of IoT devices is increasing to tens of billions, and they are creating a connected world that has the potential to make people's lives happier, safer, and more efficient. However, the introduction of smart devices, especially many mobile smart devices into IoT, will lead to many security and privacy issues, so it puts forward higher requirements for the security of mobile networks and the privacy protection of users [22]. IoT faces the internal threats caused by the security problems of devices, and many of these identical devices are not really protected and are becoming part of the so-called botnet. These small computer networks are easy to be hijacked by hackers. While most botnets traditionally consist of laptops and desktops, the growth of unsafe devices such as industrial sensors, webcams, televisions, and other smart home devices has led to a significant increase in their destructive capabilities.

Different from the distributed denial of service attacks carried out by large botnets, the internal threat of IoT lies in the operator's violation of the internal network security policy and unauthorized access to the network, so that malicious software is infected and enters in the device invading other devices of the internal network of IoT and then uses the device's own vulnerabilities to start the advanced attacks. When devices access to the network, one situation is undirected attack. Before accessing to the network, IoT devices have been infected by malware. Generally, malware attacks are used widely to infect as many devices as possible, not for a specific network. Another situation is directed attack. The attacker embeds malicious software in the IoT device, and because the specific IoT device will be directly connected to a specific internal network, it will threaten the network security, or the attacker obtains the control right of a specific device. If the user does not follow the security rules and access the device to the internal network without permission, then it can provide an opportunity for the attacker to illegally obtain sensitive information related to the internal network and then attack the whole network [23–25]. Figure 2 shows the attacker's attack model on IoT.

For the security threats existing in IoT, a whitelist can be set up. The device identification system controls the access of devices outside the list and blocks the threats at the entrance of the internal network. Meanwhile, for the devices already in the whitelist, the security management system deploys the security management measures for their communication restrictions. Although the blacklist mode can also control the access of some illegal devices, due to the complexity and diversity of the vulnerabilities of the IoT devices, it will be easier to maintain the device of the whitelist [26]. This kind of security control method for the access of whitelist devices to the network can effectively avoid the interconnection between many devices with serious loopholes and the internal network.

If a device model that does not belong to the whitelist is found, then it should be disconnected immediately to ensure the security of the intranet; if the device model belongs to the whitelist, then

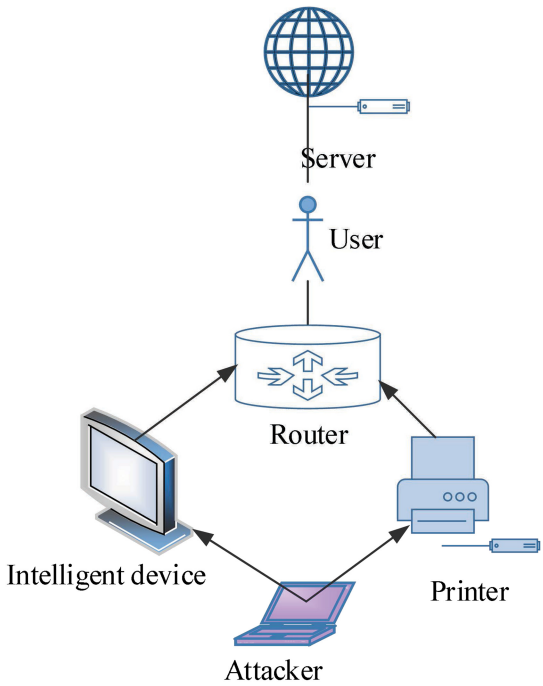

Fig. 2. Attack model of attackers on IoT.

it does not mean that there is no vulnerability risk for the device. It should be combined with the vulnerability database collected in the cloud server to analyze the unknown vulnerabilities that may exist in the device, predict the potential risks, and fully ensure the security of the internal network. Even if there is a situation that the targeted attacker embeds the device without high-risk vulnerability into the network, the traffic thinking and communication state of the device will change at this time, which plays a role in preventing the device from being identified and accessing the system [27]. If an attacker is lucky enough to invade the internal network under the undirected attack, then the security management system will continuously monitor the traffic of the internal devices of the whitelist, model based on the security threats existing in the ontology, and take corresponding security measures according to the monitoring and modeling results.

2.2.2 Threats in Network Transmission of IoT. IoT integrates the physical environment and digital environment through various communication technologies and sensor technologies and realizes the automatic identification of various entities in real life and the sharing of relevant information. At present, the system framework of IoT still continuously evolves, so the research on the security and privacy protection of IoT is still in the immature stage.

IoT network attacks are most likely to start from three aspects, namely password, network protocol, and use interface. Most of the default passwords of IoT devices have been added to the firmware, so attackers can enter the device directly. The passwords of other devices are usually weak, so they are easy to be the targets of attacks. Due to the openness of the wireless environment of IoT, attackers can obtain messages transmitted by users through wireless channels by eavesdropping and then send attacks or directly obtain some privacy information of users. IoT devices use various network protocols, and any insecure protocol will allow attackers to

steal information [28, 29]. For network interface, some IoT devices use the API interface of representational state transmission. Sensors can upload messages through the network, but any setting will give attackers the opportunity to steal private messages. Eavesdropping attack belongs to passive attack. When the user's information is eavesdropped, it is difficult for users to detect it. To deal with such attacks, sensitive or confidential information transmitted needs to be processed by users through encryption to protect the security of information transmission.

Replay attack is another attack mode of network transmission. The greatest advantage of replay attack is that it is not blocked by all firewalls, and the packets sent are legal packets. In replay attack, the attacker first intercepts the messages sent by the user by wireless or wired means through eavesdropping. Then, some of the messages are repeatedly sent to the target objects, misleading them to think that the information comes from the legitimate users, and then obtain illegal benefits. Replay attacks repeatedly power on and off the device, and observe the packet changes. If the packets obtained from multiple power on and off are the same, then the conditions for replay attacks are fully met. Just copy the contents of the packet according to the hexadecimal characters, and then quickly send massive requests to the corresponding port of the server aided by tools. This kind of attack can be resisted by adding time stamp or random value.

The communication interface allows the device to communicate with the sensor network, cloud background, mobile device APP, and other devices. The attack surface may be the firmware or driver code of the underlying communication. In general, there are two modes of the middle man attack: bypass and serial connection. The attacker is in the middle of the link at both ends of the communication and acts as the role of data exchange. The attacker can obtain the user authentication information and device control information, and then obtain the control right of the device through replay or wireless relay. For example, through man in the middle attack to decrypt HTTPS data, sensitive information can be acquired. There are some known security problems in wireless network communication interface. Regarding attack, it can form attack or even physical damage to wireless chip, DOS, security verification bypass, or code execution [30]. There are some underlying TCP/IP communication vulnerabilities, hardware implementation vulnerabilities, and other attack vectors in Ethernet device interfaces such as WiFi interface. Wireless communication Bluetooth (and BLE), ZigBee, Zwave, NFC, RFID, LoRa, and wireless Hart. There are many kinds of IoT terminal devices, various application scenarios, and various communication methods. In the process of continuous change, this is the weakest and most difficult problem for IoT security. The security mechanism can be built in to increase the difficulty of vulnerability utilization. Manufacturers can push updates to users through incremental patches, and users need to update the firmware in time.

2.3 Security Protection System in IoT

2.3.1 Remote User Authentication Mechanism Based on Elliptic Curve Cryptography. In general, to access resources or services on the remote server, users must first pass the authentication of the remote server before they can access them successfully. In the remote user authentication mechanism of insecure channel, the password-based authentication mechanism has been widely used. But the password-based user authentication scheme is not enough to meet the security requirements of various applications. As the smart card has the functions of storage, encryption, and calculation, the identity authentication scheme can solve the problem of server storing password table, and improve the security of the authentication scheme. A secure and effective remote user authentication scheme needs to meet the basic requirements of user anonymity, single registration center, two-way authentication, and key negotiation [31]. The security hash function is used to encrypt the secret values of users and servers, and the elliptic curve encryption is combined

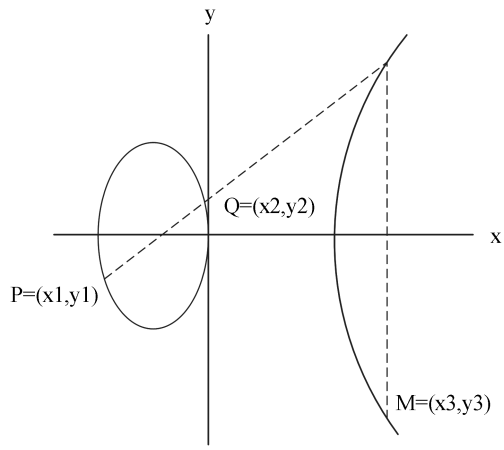

Fig. 3. Addition operation of different points on elliptic curve.

with the security hash function, so that users, servers, and registry can realize the encryption of interactive information.

In the environment of IoT, Koblitz and Miller proposed a remote user authentication scheme based on **Elliptic Curve Cryptography (ECC)** to solve the security problem of users accessing IoT through various wireless terminals. The calculation principle of RSA is simple: to get the encrypted message by power modulus, and the requirement of high security level is bound to increase the number of power bits. Hence, the length of RSA's encrypted bits is a double-edged sword, which increases the difficulty of decoding and improves the security performance and increases the calculation time. ECC algorithm is much less complex than RSA, but it gets the same security level of RSA. If a prime number p is set in the elliptic curve, and the finite field composed of module P is expressed by $GF(p)$, then the elliptic curve above is expressed as Equation (1),

$$y^2 = x^3 + ax + b \tag{1}$$

In Equation (1), $a, b \in GF(p)$.

The addition of points on the elliptic curve has its geometric significance. If points P and Q are two different points on the elliptic curve E, then make a straight line through points P and Q, which must intersect the elliptic curve at point M′, and then make a straight line perpendicular to the X axis along point M′, and the straight line intersects the elliptic curve at point M, as shown in Figure 3.

If point P = Q, then make a tangent about the elliptic curve through point P, which intersects the elliptic curve at point M′, and then make a straight line perpendicular to the X axis along point M′, the straight line intersects the elliptic curve at point M, as shown in Figure 4.

Supposing that $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ are the points on the elliptic curve, then the coordinates of the point $M = (x_3, y_3)$ in the ordinary addition operation of the different points on the elliptic curve can be expressed as Equations (2) and (3),

$$x_3 = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)^2 - x_1 - x_2, \tag{2}$$

$$y_3 = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x_1 - x_3) - y_1. \tag{3}$$

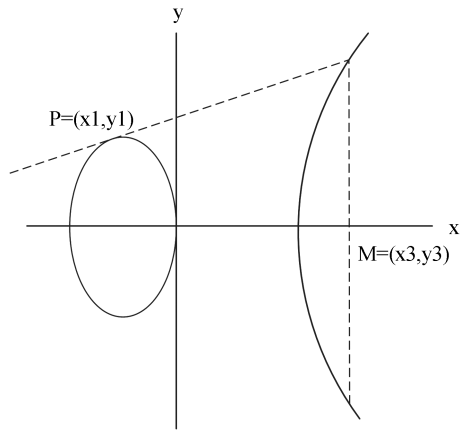

Fig. 4. Multiplication of points on elliptic curve

The coordinates of the points in the multiplication operation of the points on the elliptic curve can be expressed as Equations (4) and (5),

$$x_3 = \left(\frac{3x_1^2 + a}{2y_1}\right)^2 - 2x_1, \tag{4}$$

$$y_3 = \left(\frac{3x_1^2 + a}{2y_1}\right)(x_1 - x_3) - y_1. \tag{5}$$

If a website stores password in plain text instead of hash, then it will seriously threaten security. If someone invades the website database and gets the mailbox and password of plain text login, then he or she can try the same login credentials on any other website. Hash function, as an important part of information security field, can safely store passwords, find duplicate records, and quickly store and retrieve data [32]. Determinacy is very essential for the secure storage password. It can change the original simple password into a long string of hash characters by disturbing and disrupting the plaintext password set by the user. If someone sees the encrypted version password, then he or she will not know the original password, which is very important. It means that, as a website developer, only the hash value of the user's password needs to be stored for verification. Figure 5 shows the specific representation of hash function. When the user registers, the developer hashes the password and stores it in the database; when the user logs in, the developer hashes their input, and then compares the hash values in the database—because the same input always produces the same hash value, which is effective.

2.3.2 Key Management Scheme in Wireless Sensor. WSN has been widely used, and meanwhile, the related security issues have also attracted people's attention, in which the orderly management of key is the basis to ensure the security of the system. The main purpose of key management is to prevent the key from leaking from abnormal channels, and it is also an effective way to change the key regularly. A relatively secure key management system can store all the keys (including some sensitive configuration files) on a server and provide the API to obtain the key through web service. When each Web application needs to use the key, it requests the key management system through the API with authentication information to obtain the key dynamically. Web application cannot write the key to the local file but can only load it into memory, so dynamic key acquisition

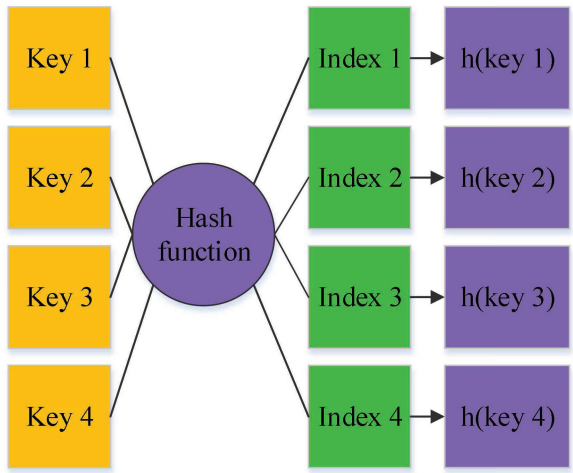

Fig. 5. Specific representation of Hash function.

can protect the privacy of the key to the greatest extent. Centralized key management reduces the coupling of the system to the key and benefits the regular replacement of the key.

The common WSN key pre-distribution schemes are mainly based on key distribution center, key pre-distribution, and multi-key distribution based on grouping and clustering. Among them, the distribution of group and cluster is more suitable for practical applications, which can store the key in a decentralized way, thus reducing the storage consumption of the key. To crack a key space, the attackers must obtain t+1 nodes in the area. When the network node finds that the sensor node has been captured, it will send the updated message to the nodes in the area. After receiving the key update message, the nodes not captured can obtain the new key information through calculation [33].

In initialization, the server constructs a Vandermonde matrix G in the finite field GF (q), where t is the security threshold of the shared key. Take a region C_i as an example, where there are m nodes in total, and each node has a unique ID identifier. As the seed of Vandermonde matrix G, G_i indicates the Vandermonde matrix of C_i, and then, Equation (6) is obtained,

$$G_i = \begin{bmatrix} 1 & 1 & \dots & 1 \\ (\mathrm{ID}_{\mathrm{m_{i1}}})^1 & (\mathrm{ID}_{\mathrm{m_{i2}}})^1 & \dots & (ID_{m_{iN}})^1 \\ (ID_{m_{i1}})^2 & (ID_{m_{i2}})^2 & \dots & (ID_{m_{iN}})^2 \\ \dots & & & \\ (ID_{m_{i1}})^t & (ID_{m_{i2}})^t & \dots & (ID_{m_{iN}})^t \end{bmatrix}. \tag{6}$$

The server randomly generates a $(t+1) \times (t+1)$ symmetric secret matrix D_i on the finite field GF (q), and the Blom matrix of the region C_i is expressed as Equation (7),

$$A_i = (D_i \cdot G_i)^T. \tag{7}$$

The key space is matrix K, which represents the ith key space:

$$K_i = A_i \cdot G_i = (D_i \cdot G_i)^T \cdot G_i = {G_i}^T \cdot D_i \cdot G_i = (A_i \cdot G_i)^T = K_i^T. \tag{8}$$

The key information stored by all nodes in the regular hexagon grid C_i comes from the matrixes G_i and A_i. The nodes m_{ij} and m_{ik} in the region C_i store information $\{ID_{m_{ij}}, row_j(A_i)\}$ and $\{ID_{m_{ik}}, row_j(A_i)\}$, respectively. The shared key space between the two nodes can be expressed as

Equations (9)–(11),

$$K_m = (A_m \cdot G_i), \tag{9}$$

$$col(G_i) = [1 \quad (ID_{m_{ik}})^1 \quad \cdots \quad (ID_{m_{ik}})^t]^T, \tag{10}$$

$$K_{m_{ij}m_{ik}} = [row_j(A_i)] \times [col_k(G_i)]. \tag{11}$$

From symmetry, Equation (12) is obtained,

$$K_{m_{ij}m_{ik}} = K_{m_{ik}m_{ij}}. \tag{12}$$

The session key $K_{m_{ij}m_{ik}}$ of the nodes m_{ij} and m_{ik} can be obtained.

When a node in the WSN is detected to be captured by the enemy, the system can revoke the established session key. On the finite domain GF (q), when the common node m_{ij} receives the key update message, it calculates according to the information stored by itself:

$$K_{m_{ij}m_{ik}}{}^* = [col_j(G_i)][row_j(A_i)]. \tag{13}$$

Determine whether $K_{m_{ij}m_{ik}}{}^* = K_{m_{ik}m_{ij}}$ is true or not. If it is false, then the message is not sent from the gateway node, then discard the message. Otherwise, the key is updated,

$$[[row_j(H_i)]K_{m_{ij}m_{ic}}]K_{m_{ij}m_{ic}} = [row_j(H_i)], \tag{14}$$

$$[row_j(A_i{}^*)] = [row_j(H_i)] - [row_j(A_i)]. \tag{15}$$

Get new key information $row_j(A_i{}^*)$. For compromised nodes, $row_j(H_i)$ is 0, key update cannot be performed, and data communication after WSN cannot be performed.

2.4 LoRa Communication and Signal Source Identification

2.4.1 LoRa Wireless Communication Transmission. As the demand of intelligent city and large-scale IoT technology increases, the traditional IoT communication technology has been unable to meet the existing application requirements. LoRa, as one of the specific implementation methods of low-power wide area network, has been deployed in more fields, such as target location and track tracking due to mature protocol architecture. The most prominent feature of LoRa is long distance, which breaks through the coverage scenarios that need relay to solve before, or requires low power consumption and battery power supply. As the network deployment increases, LoRa signal sources also increase. To ensure that LoRa conflict signal can correctly decode, track, and identify information, and prevent malicious signal source attacks from threatening network security, LoRa, a new wireless communication technology, is explored to ensure the signal security of IoT. LoRaWAN is an asynchronous protocol based on ALOHA, that is, nodes can sleep long or short according to specific application scenario requirements, and LoRaWAN has 3–5 times power consumption advantage over other cellular technologies.

LoRa receiver has the RX continuous mode and the RX single mode. The RX continuous mode is kept in reception until manual switching, while the RX single mode receives only one reception and automatically switches to standby mode. In the RX single mode, the preamble is searched in the given time window. After finding the pilot code, the modem starts to receive the packet and then switches it to standby mode after receiving it. If the pilot code is not found within the time window, then the chip will be switched to standby mode. Because the time when the packet reaches the time window is unknown in most scenarios, RX continuous mode should be adopted. The modem will scan the channel continuously. Once the preamble is detected, it will detect the preamble, then receive the packet, and continue to detect the next preamble. Compared with other high-level wireless chips, LoRa can improve the receiving sensitivity by 20 dB in the field of IoT.

In addition, the power consumption of LoRa is very low, and due to the application of LFM spread spectrum modulation, it can effectively reduce interference.

2.4.2 LoRa Signal Source Recognition Based on Multi-Features. Signal features include amplitude, IMF amplitude, and marginal spectrum. These features can form a feature set to describe the characteristics of corresponding signal sources in different dimensions. According to different scenes and acquired signal features, select appropriate learning algorithm to build attribute value model for these features. Once signal features are input, corresponding signal source categories can be found from the model according to attribute values. The signal features come from the signal itself, but the signal will be affected by the noise and transmission environment, which makes the internal relationship of the signal features more difficult to capture. To identify the signal source accurately, three classifiers: **random forest (RF), support vector machine (SVM)**, and **artificial neural network (ANN)** are selected to process the signal feature set.

Random forest first constructs a decision tree and then continues to construct multiple decision trees. It combines multiple decision trees by self-service sampling method and selects attribute sampling in each decision tree learning. As a method of predicting variables in machine learning tree algorithm, the random vector value obtained by independent sampling plays a decisive role in every tree with the same distribution of RF. As a supervised learning method, SVM uses an optimal linear hyperplane to partition data and define the boundary between the two types of data samples. The kernel function in SVM can map the non-linear classification to the high-dimensional space, so that the problems that cannot be classified originally are solved by finding the linear hyperplane. ANN can use the neural network to deal with the classification work of huge training set and complex internal relationship of feature set, and the anti-interference and associative memory function of ANN also has advantages. When using multi-features for signal source recognition, it is necessary to combine different features, select the signal with the highest recognition accuracy according to the actual problems, and input it into the classifier as the classification basis.

2.4.3 Backoff Algorithm in Lora Multipoint Communication. Backoff algorithm aims to ensure that all nodes in the network send data fairly. A good backoff algorithm can reduce conflict and delay and improve throughput and network performance. The most typical backoff algorithm is the **binary exponential backoff algorithm (BEB)**. The greatest problem of the BEB algorithm is that when the data are sent successfully, it directly resets the competitive window to the minimum value, that is, it always gives the maximum competitive priority to the node that sent successfully the last time, which may lead to the unfair phenomenon of the network. And even the channel has been occupied by a certain node that other nodes cannot compete to the channel. On this basis, the BEB algorithm is optimized and the new backoff algorithm is named as REBEB. The main idea is to reduce the contention window when communication conflicts.

After the general industrial networking equipment is determined, the network is basically formed, the number of nodes in the network will not change frequently, and the production information and data volume of equipment will be different due to the production and technology of enterprises, so the network competition degree will maintain stably. When the communication is successful, the continuous communication success times are judged. Therefore, the factors that affect the performance of the backoff algorithm are the number of nodes in the network, the initial competition window, and the maximum and minimum competition window.

2.5 Simulation Experiment

To test the indoor penetration ability of LoRa, different floors of the same building are selected for testing. The floor is 13 floors, and the building height is 52 m. Three nodes, A, B and C, are

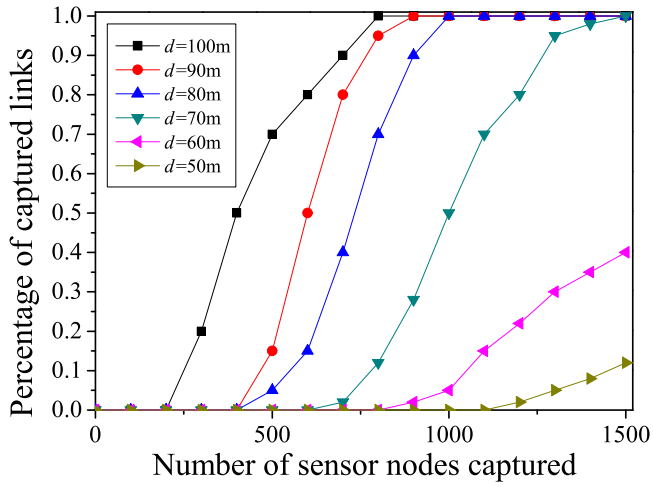

Fig. 6. The relationship between the number of sensor nodes captured and the impact of the link.

selected for the test scheme. Each node is tested alternately, and the **spread-spectrum factor (SF)** is 7–12. In the test process, different test points open three nodes respectively to transmit data to the gateway. After forwarding the data to the server, the server can directly collect information, read the count value, and know the lost count, thus obtaining the packet loss rate.

A network can judge its performance by normalized throughput, fairness, average delay, packet loss rate, and other indicators. The normalized throughput and fairness of the network are analyzed by mathematical methods. The simulation of LoRa communication protocol is realized by MATLAB, and the simulation value is compared with the theoretical value to analyze the performance of LoRa communication protocol. Set up a star network topology environment, including N LoRa nodes (responsible for reporting data to the hub) and 1 LoRa hub. N nodes must communicate with the hub to form a competitive network. In the simulation experiment, the failure of information transmission caused by collision is ignored, and the network is in saturated state. The scenarios of 5, 10, 15, 20, 25, and 30 nodes are simulated, respectively.

The normalized throughput analysis of IoT communication system includes theBEB and new back-off algorithm (REBEB). In the actual application scenario, the protocol layer fixes the parameters such as the competition window and the maximum allowed back-off times. It is known that after the maximum allowed back-off times, BEB will not greatly affect the system performance.

3 RESULTS AND DISCUSSION

3.1 Security Analysis of IoT Communication Chain

A secure gateway node is used as the root CA to manage the certificate and key of the gateway node and cluster head node. The idea of certificate-based three-way authentication can resist replay attack. Figure 6 shows the relationship between the number of sensor nodes captured and the impact of the link. Based on the cluster structure of WSN, it is divided into non-overlapping hexagon network areas for regional management, and d represents the side length of the regular hexagon area. Figure 6 suggests that, as the area range is expanded, when the number of sensor nodes captured is the same, the proportion of communication links affected increases gradually. It means that the captured nodes are more concentrated in a region, so the region is more vulnerable to attack and cracking.

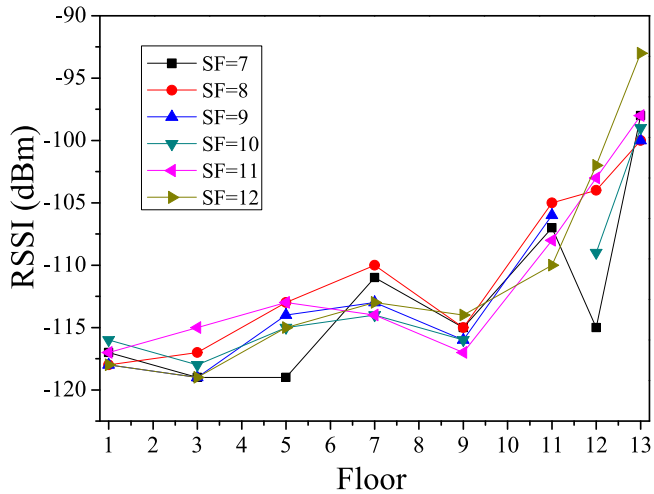

Fig. 7. The relationship between floor and receiving sensitivity under different SF conditions.

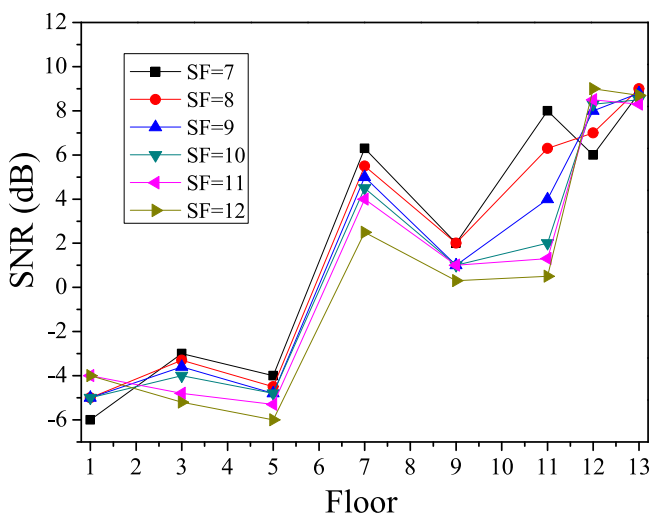

Fig. 8. The relationship between floors and SNR under different SF conditions.

3.2 Performance Analysis of LoRa Communication System

According to the test results of performance of LoRa communication system, the relationship between floors and receiving sensitivity under different SF conditions is shown in Figure 7, the relationship between floors and **signal-to-noise ratio (SNR)** under different SF conditions is shown in Figure 8, and the relationship between floors and receiving rate under different SF conditions is shown in Figure 9. Figures 7 to 9 show that SF obviously impacts the transmission of LoRa communication signal in the floor. Figure 7 suggests that with the increase of SF, the impact on SNR increases significantly, the sensitivity of the received signal increases, and the increase of floor interval will weaken the sensitivity of the received signal. Figure 8 shows that SNR will decrease with the decrease of floors, and the sensitivity limit of acceptable signal reception will also decrease with the increase of SF. Figure 9 shows that when SF is 12, there is basically no packet loss;

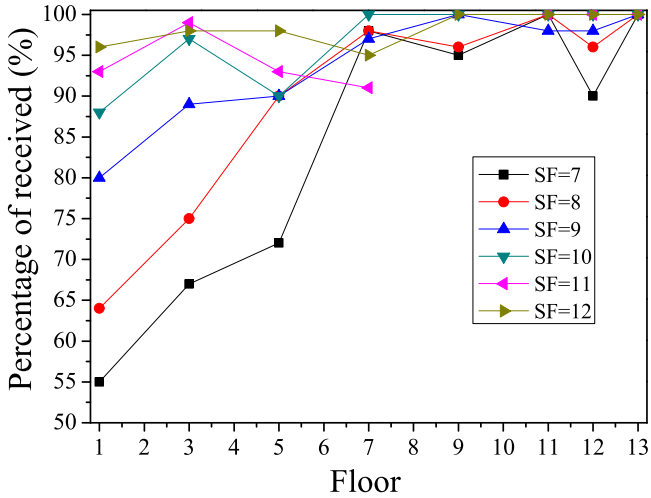

Fig. 9. The relationship between the floor and the packet loss rate under different SF conditions.

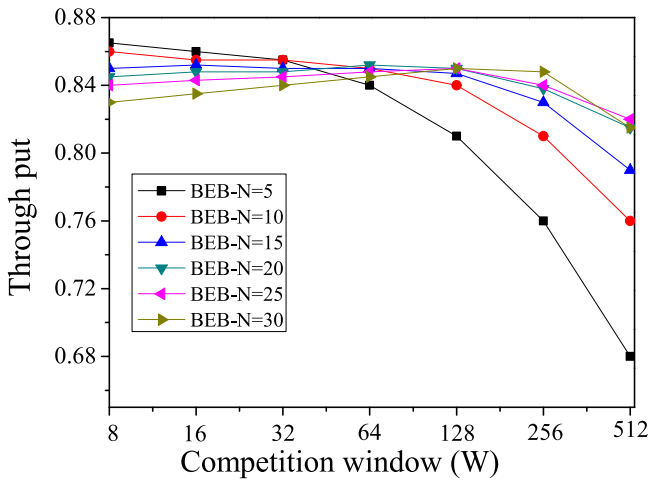

Fig. 10. Comparison of throughput of different nodes and competition windows (BEB).

when SF is 7, the packet loss rate of the first floor is about 55%. With the increase of the floor, the number of packet loss of equipment is also increasing. When SF is increased, the terminal device can receive information effectively at a lower SNR.

To summarize, the increase of SF can improve the signal receiving ability of terminal equipment and ensure the accurate transmission of information under low SNR. And when SF is low, the growth rate of packet loss rate will gradually accelerate.

3.3 Security Performance of Wireless Communication Protocol For IoT Devices

The throughput of different nodes and different competition windows are compared, as shown in Figure 10. It shows that the normalized throughput of BEB algorithm is significantly different in different node numbers and different window initial values. When the number of nodes is small (N = 5–10), the throughput of W value in 16–32 is better. When N increases (N = 25–30), the

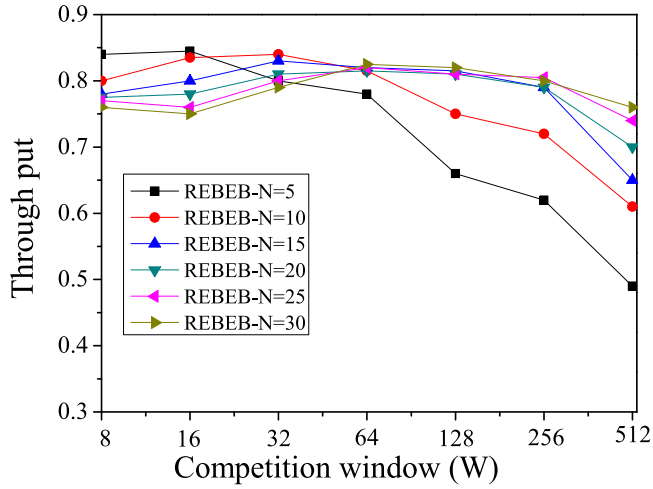

Fig. 11. Comparison of throughput of different nodes and competition windows (REBEB).

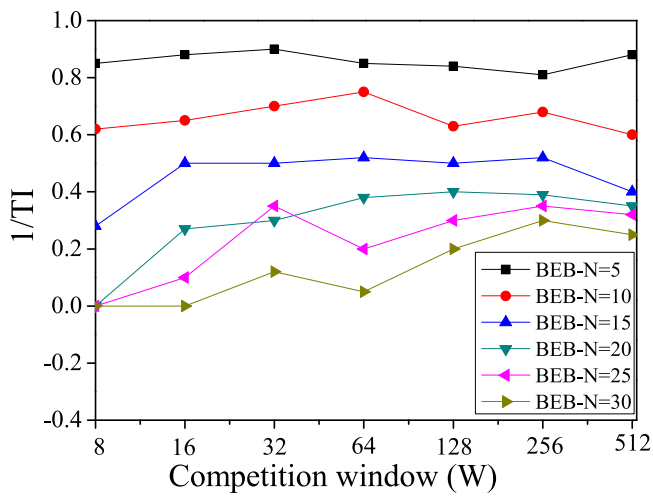

Fig. 12. Security analysis of different nodes and different competition windows (BEB).

throughput of W is better in 64–128. When the competition window is small, there will be more conflicts, which will reduce the exchange rate of effective data and the throughput.

The throughput of different nodes and different contention windows in REBEB algorithm are compared, as shown in Figure 11. REBEB is basically similar to the BEB regarding throughput and N and W values. REBEB pays more attention to fairness. When a node successfully transmits several times in a row, it will actively expand the competition window and give more channel acquisition rights to other nodes. In the network with different number of nodes, both BEB and REBEB have their corresponding optimal value of competition window.

Next, the security of the IoT communication system is analyzed.The security of different nodes and different competition windows under BEB and REBEB is analyzed. as shown in Figures 12 and 13. It shows that the fairness of BEB is greatly affected by N that the more the nodes are, the worse the security is. When N is greater than 15, the security almost zero. The security of REBEB is more

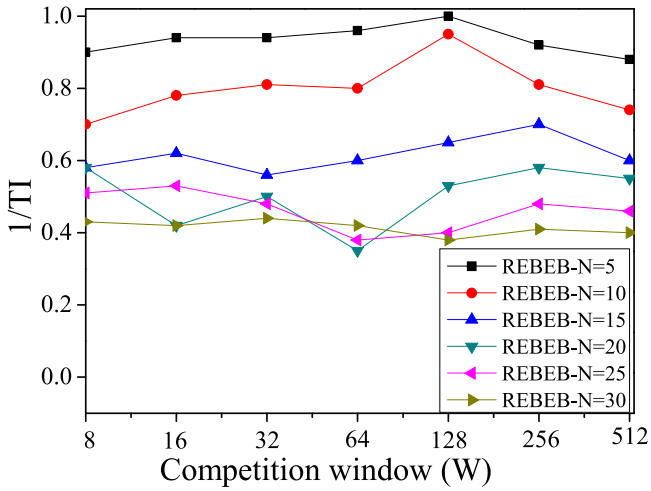

Fig. 13. Security analysis of different nodes and competition windows (REBEB).

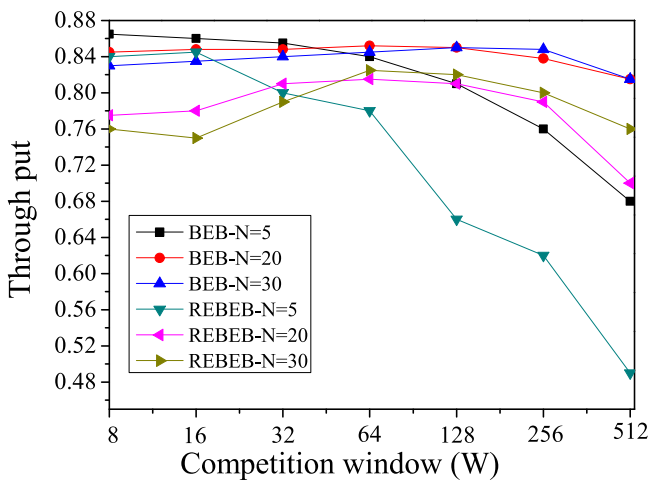

Fig. 14. Fairness comparison of two algorithms when the number of nodes is 5, 20, and 30.

than 0.4 in different nodes and competition windows, and the security of the system is better when the number of nodes is small.

The fairness and security of the two back-off algorithms are compared in different number of nodes, and the results are shown in Figures 14 and 15. In different number of nodes, the fairness of the new back-off algorithm REBEB is significantly better than that of BEB, and the highest fairness of REBEB is 0.985.

Due to binary exponential backoff algorithm, when the number of nodes is large, the more communication conflicts there are in the network, and the competition window is doubled after node conflicts, the more time the nodes spend waiting for the next access channel. However, a node that successfully acquires the channel will reduce its competition window to the minimum value, that is, the time that the node waits for the next access channel is very short, so the probability of the node successfully accessing the channel again will be greatly increased. As a result, the conflicted nodes are more difficult to access the channel, and the successful nodes are more likely

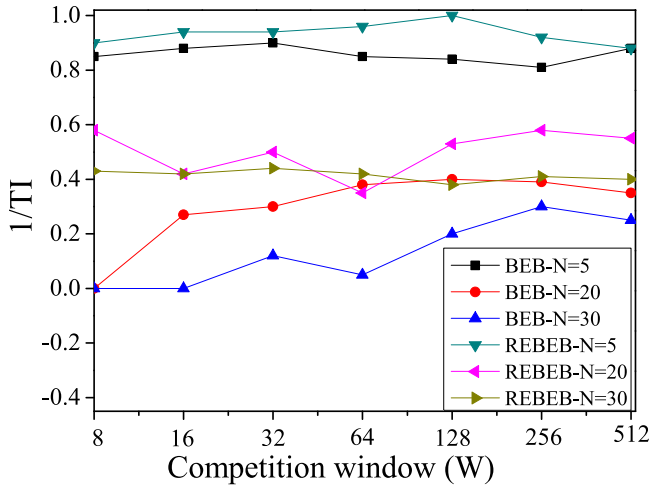

Fig. 15. Security comparison of two algorithms when the number of nodes is 5, 20, and 30.

to access the channel unfairly. Therefore, the BEB algorithm has poor fairness performance when the number of nodes is large.

4 CONCLUSIONS

The construction of smart city is a complex engineering, in which IoT technology as a key technology, makes information acquisition and environment monitoring through sensors, RFID tags, and other equipment for infrastructure and buildings. As the IoT system usually needs to use wireless communication with open characteristics, which brings convenience to users, but also threats the security of the system and the privacy of users. Based on the security requirements of IoT, this study analyzes the security of different IoT application technologies. Considering that with the increase of network deployment, LoRa signal sources also increase, to ensure that LoRa conflict signal can correctly track and identify information, and prevent malicious signal source attacks from threatening network security, LoRa, a new wireless communication technology, is discussed to ensure the signal security of IoT.

It is found that, the setting of time slot interval based on LoRa, the fairness of REBEB, a new back-off algorithm, is significantly better than that of BEB in the network with different number of nodes. Moreover, to a certain extent, its security is significantly improved, and the REBEB algorithm basically meets the communication requirements from the throughput protocol. In conclusion, the IoT system based on LoRa communication technology can effectively improve the security performance of the system in the construction of smart city, and avoid the security threats in the IoT signal transmission. The analysis of signal feature extraction is not deep enough. At present, the extracted features are still based on time-frequency feature analysis, but the existing signal processing methods are very rich. Therefore, in the next studies, the processing methods of signal characteristics and internal relations will be reflected to improve the reliability of signal characteristics. In future, this technology will be improved by Big data and 6G technology [34–43].

REFERENCES

[1] E. F. Z. Santana, A. P. Chaves, M. A. Gerosa, et al. 2017. Software platforms for smart cities: Concepts, requirements, challenges, and a unified reference architecture. *ACM Comput. Surv.* 50, 6 (2017), 1–37.

[2] Q. Zhu, S. W. Loke, R. Trujillo-Rasua, et al. 2019. Applications of distributed ledger technologies to the internet of things: A survey. *ACM Comput. Surv.* 52, 6 (2019), 1–34.

[3] L. Gonzalez-Manzano, J. M. D. Fuentes, and A. Ribagorda. 2019. Leveraging user-related internet of things for continuous authentication: A survey. *ACM Comput. Surv.* 52, 3 (2019), 1–38.
[4] R. Khatoun and S. Zeadally. 2016. Smart cities: Concepts, architectures, research opportunities. *Commun. ACM* 59, 8 (2016), 46–57.
[5] J. Liu, H. Shen, H. S. Narman, et al. 2018. A survey of mobile crowdsensing techniques: A critical component for the internet of things. *ACM Trans. Cyber-Phys. Syst.* 2, 3 (2018), 1–26.
[6] A. Bennaceur, T. T. Tun, A. K. Bandara, et al. 2018. Feature-driven mediator synthesis: Supporting collaborative security in the internet of things. *ACM Trans. Cyber-Phys. Syst.* 2, 3 (2018), 1–25.
[7] A. Rullo, D. Midi, E. Serra, et al. 2017. Pareto optimal security resource allocation for Internet of Things. *ACM Trans. Priv. Secur* 20, 4 (2017), 1–30.
[8] N. Kolbe, S. Kubler, J. Robert, et al. 2019. Linked vocabulary recommendation tools for internet of things: a survey. *ACM Comput. Surv.* 51, 6 (2019), 1–31.
[9] O. Novo and M. D. Francesco. 2020. Semantic interoperability in the IoT: Extending the web of things architecture. *ACM Trans. IoT* 1, 1 (2020), 1–25.
[10] S. Siboni, A. Shabtai, N. O. Tippenhauer, et al. 2016. Advanced security testbed framework for wearable IoT devices. *ACM Trans. Internet Technol.* 16, 4 (2016), 1–25.
[11] Y. Mehmood, F. Ahmad, I. Yaqoob, et al. 2017. Internet-of-things-based smart cities: Recent advances and challenges. *IEEE Commun. Mag.* 55, 9 (2017), 16–24.
[12] A. Gharaibeh, M. A. Salahuddin, S. J. Hussini, et al. 2017. Smart cities: A survey on data management, security, and enabling technologies. *IEEE Commun. Surv. Tutor.* 19, 4 (2017), 2456–2501.
[13] B. A. Mozzaquatro, R. Jardim-Goncalves, and C. Agostinho. 2015. Towards a reference ontology for security in the internet of things. In *Proceedings of the IEEE International Workshop on Measurement & Networking*. IEEE, 2015, 289–296.
[14] W. Li, H. Song, and F. Zeng. 2017. Policy-based secure and trustworthy sensing for internet of things in smart cities. *IEEE IoT J.* 5, 2 (2017), 716–723.
[15] D. Minoli, K. Sohraby, and B. Occhiogrosso. 2017. IoT considerations, requirements, and architectures for smart buildings—Energy optimization and next-generation building management systems. *IEEE IoT J.* 4, 1 (2017), 269–283.
[16] J. Lin, W. Yu, N. Zhang, et al. 2017. A survey on internet of things: Architecture, enabling technologies, security and privacy, and applications. *IEEE IoT J.* 4, 5 (2017), 1125–1142.
[17] M. Stolpe. 2016. The internet of things: Opportunities and challenges for distributed data analysis. *ACM SIGKDD Explor. Newslett.* 18, 1 (2016), 15–34.
[18] M. C. Schraefel, R. Gomer, A. Alan, et al. 2017. The internet of things: interaction challenges to meaningful consent at scale. *Interactions* 24, 6 (2017), 26–33.
[19] M. E. Jennex. 2017. Big data, the internet of things, and the revised knowledge pyramid. *ACM SIGMIS Database* 48, 4 (2017), 69–79.
[20] B. Billet and V. Issarny. 2017. Spinel: An opportunistic proxy for connecting sensors to the Internet of Things. *ACM Trans. Internet Technol.* 17, 2 (2017), 1–21.
[21] M. Ambrosin, P. Braca, M. Conti, et al. 2017. Odin: O bfuscation-based privacy-preserving consensus algorithm for decentralized information fusion in smart device networks. *ACM Trans. Internet Technol.* 18, 1 (2017), 1–22.
[22] A. Makhoul, C. Guyeux, M. Hakem, et al. 2016. Using an epidemiological approach to maximize data survival in the internet of things. *ACM Trans. Internet Technol.* 16, 1 (2016), 1–15.
[23] J. Dizdarević, F. Carpio, A. Jukan, et al. 2019. A survey of communication protocols for internet of things and related challenges of fog and cloud computing integration. *ACM Comput. Surv.* 51, 6 (2019), 1–29.
[24] M. Centenaro, L. Vangelista, A. Zanella, et al. 2016. Long-range communications in unlicensed bands: The rising stars in the IoT and smart city scenarios. *IEEE Wireless Commun.* 23, 5 (2016), 60–67.
[25] Y. Qian, D. Wu, W. Bao, et al. 2019. The internet of things for smart cities: Technologies and applications. *IEEE Netw.* 33, 2 (2019), 4–5.
[26] Y. Zhang, Z. Xiong, D. Niyato, et al. 2020. Information trading in internet of things for smart cities: A market-oriented analysis. *IEEE Netw.* 34, 1 (2020), 122–129.
[27] E. Fernandes, A. Rahmati, K. Eykholt, et al. 2017. Internet of things security research: A rehash of old ideas or new intellectual challenges. *IEEE Secur. Priv.* 15, 4 (2017), 79–84.
[28] J. Hu, K. Yang, S. T. Marin, et al. 2018. Guest editorial special issue on Internet-of-Things for smart cities. *IEEE IoT J.* 5, 2 (2018), 468–472.
[29] K. F. Tsang and V. Huang. 2019. Conference on sensors and internet of things standard for smart city and inauguration of IEEE P2668 internet of things maturity index [Chapter News]. *IEEE Industr. Electr. Mag.* 13, 4 (2019), 130–131.
[30] H. Fu, M. Wang, P. Li, et al. 2019. Tracing knowledge development trajectories of the internet of things domain: A main path analysis. *IEEE Trans. Industr. Inf* 15, 12 (2019), 6531–6540.

[31] H. Wang, S. Yu, S. Zeadally, et al. 2020. Introduction to the special section on network science for Internet of Things (IoT). *IEEE Trans. Netw. Sci. Eng.* 7, 1 (2020), 237–238.
[32] M. Sookhak, H. Tang, Y. He, et al. 2018. Security and privacy of smart cities: A survey, research issues and challenges. *IEEE Commun. Surv. Tutor.* 21, 2 (2018), 1718–1743.
[33] Y. Lu and L. Da Xu. 2018. Internet of things (iot) cybersecurity research: A review of current research topics. *IEEE IoT J.* 6, 2 (2018), 2103–2115.
[34] Z. Lv, S. Zhang, and W. Xiu. 2020. Solving the security problem of intelligent transportation system with deep learning. *IEEE Trans. Intell. Transport. Syst.* DOI: 10.1109/TITS.2020.2980864
[35] F. Piccialli, G. Casolla, S. Cuomo, F. Giampaolo, and V. S. di Cola. 2020. Decision making in IoT environment through unsupervised learning. *IEEE Intelligent Systems.* 35, 1 (2020), 27–35.
[36] Z. Lv and H. Song. 2020. Mobile internet of things under data physical fusion technology. *IEEE IoT J.* 7, 5 (2020), 4616–4624.
[37] G. Casolla, S. Cuomo, V. S. Di Cola, and F. Piccialli. 2020. Exploring Unsupervised Learning techniques for the Internet of Things. *IEEE Trans. Industr. Inf.* 16, 4 (2020), 2621–2628.
[38] R. Shi, Y. Gan, and Y. Wang. 2018. Evaluating scalability bottlenecks by workload extrapolation. In *Proceedings of the IEEE 26th International Symposium on Modeling, Analysis, and Simulation of Computer and Telecommunication Systems (MASCOTS'18).* 333–347.
[39] Z. Lv, W. Kong, X. Zhang, D. Jiang, H. Lv, and X. Lu. 2020. Intelligent security planning for regional distributed energy internet. *IEEE Trans. Industr. Inf.* 16, 5 (2020), 3540–3547.
[40] F. Piccialli, S. Cuomo, V. S. di Cola, and G. Casolla. 2019. A machine learning approach for IoT cultural data. *J. Ambient Intell. Human. Comput.* (2019), 1–12.
[41] Z. Lv, X. Li, H. Lv, and W. Xiu. 2020. BIM Big Data Storage in WebVRGIS. *IEEE Trans. Industr. Inf.* 16, 4 (2020), 2566–2573.
[42] S. Dang, O. Amin, B. Shihada, and M. S. Alouini. 2020. What should 6G be?. *Nat. Electr.* 3, 1 (2020), 20–29.
[43] H. Lu, Y. Zhang, Y. Li, C. Jiang and H. Abbas. 2021. User-Oriented virtual mobile network resource management for vehicle communications. *IEEE Trans. Intell. Transport. Syst.* 22, 6 (2021), 3521–3532.

Received April 2020; revised June 2020; accepted June 2020